(i) For any $x \in \mathbb{R}$, we have $|x| = |-x|$ and thus, $|x| = |y|$ if and only if $x = y$ or $x = -y$.
(ii) $|x - a| = r$ if and only if $r \geq 0$ and $x - a = r$ or $x - a = -r$.

### 2.3.2 Equations Involving Absolute Value

Note that *a real number $a$ is said to be a **solution** of an equation or an inequality, if the statement obtained after replacing the variable by $a$ is true.*

Next we shall learn solving equations involving absolute value.

**Example 2.1** Solve $|2x - 17| = 3$ for $x$.

**Solution:**
$|2x - 17| = 3$. Then, we have $2x - 17 = \pm 3$ which implies $x = 10$ or $x = 7$.

**Example 2.2** Solve $3|x - 2| + 7 = 19$ for $x$.

**Solution:**
$3|x - 2| + 7 = 19$. So that we have, $|x - 2| = \frac{19-7}{3} = 4$.
Thus, we have either $x - 2 = 4$ or $x - 2 = -4$.
Therefore the solutions are $x = -2$ and $x = 6$.

**Example 2.3** Solve $|2x - 3| = |x - 5|$.

**Solution:**
We know that $|u| = |v|$ if and only if $u = v$ or $u = -v$.
Therefore, $|2x - 3| = |x - 5|$ implies $2x - 3 = x - 5$ or $2x - 3 = 5 - x$.
Solving these two equations, we get $x = -2$ and $x = \dfrac{8}{3}$.
Hence, both $x = -2$ and $x = \dfrac{8}{3}$ are solutions.

### 2.3.3 Some Results For Absolute Value

(i) If $x, y \in \mathbb{R}$, $|y + x| = |x - y|$, then $xy = 0$.
(ii) For any $x, y \in \mathbb{R}$, $|xy| = |x||y|$.
(iii) $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$, for all $x, y \in \mathbb{R}$ and $y \neq 0$.
(iv) For any $x, y \in \mathbb{R}$, $|x + y| \leq |x| + |y|$.

### 2.3.4 Inequalities Involving Absolute Value

Here we shall learn to solve inequalities involving absolute values. First we analyze very simple inequalities such as (i) $|x| < r$ and (ii) $|x| > r$.

(i) Let us prove that $|x| < r$ if and only if $-r < x < r$. Note that $r > 0$ as $|x| \geq 0$.
There are two possibilities to consider depending on the sign of $x$.

Case (1): If $x \geq 0$, then $|x| = x$, so $|x| < r$ implies $x < r$.
Case (2): If $x < 0$, then $|x| = -x$, so $|x| < r$ implies $-x < r$ that is, $x > -r$.

Therefore we have, $|x| < r$ if and only if $-r < x < r$, that is $x \in (-r, r)$.